Efficient QLoRA Fine-Tuning of Qwen2.5-VL for Multi-Class Tomato Disease Classification and Cross-Domain Evaluation

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Abstract. Vision-language models provide a flexible foundation for fine-grained image understanding, but adapting multi-billion-parameter models to specialized agricultural classes remains computationally expensive and vulnerable to class imbalance and domain shift. This work evaluates parameter-efficient fine-tuning of Qwen2.5-VL-7B-Instruct for tomato disease classification using the private tomato image dataset. From 7,095 raw images, an integrity audit with image verification, SHA-256 exact deduplication, and removal of contradictory cross-class duplicates retained 6,545 images across 11 classes, divided deterministically into 5,247 training, 649 validation, and 649 test images. The model was adapted with QLoRA using 4-bit NF4 double quantization, BF16 computation, language-projection LoRA with rank 64 and alpha 128, a frozen vision tower, a 532-pixel patch-aligned maximum image side, and inverse-square-root class-balanced sampling. Prompt tokens were masked so that optimization supervised only the target class label. Training on a single NVIDIA RTX 4500 Ada 24-GB GPU completed 1,908 optimizer steps (2.909 epochs) in 192.7 min with 10.57 GB peak reserved memory. On the internal test set, the selected adapter achieved 92.76% accuracy, 93.78% macro F1, 95.57% macro recall, and 0.327 s/image latency. An independent 176-image, 10-class benchmark yielded 69.32% accuracy and 71.95% macro F1, exposing a substantial cross-domain generalization gap, particularly for Late Blight. The results show that QLoRA can efficiently adapt a 7B vision-language model for high in-domain tomato disease classification accuracy on a single workstation GPU while highlighting the need for broader domain coverage for deployment.

Keywords: Tomato disease classification · Vision-language model · Qwen2.5-VL · QLoRA · Parameter-efficient fine-tuning · Domain shift

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1 Introduction

Image-based plant disease recognition has become a major application of deep learning because visually similar symptoms can complicate manual diagnosis and delay intervention. Early CNN studies demonstrated strong disease discrimination from labelled leaf images [1, 2], while later work evaluated deeper networks, transfer learning, and broader plant-disease collections [3, 4]. Reviews emphasize that reported performance depends strongly on dataset composition, acquisition conditions, preprocessing, and evaluation design [5].

Tomato disease recognition is particularly challenging because disease and pest symptoms can overlap visually. Prior work has studied real-time tomato disease and pest recognition [6], tomato disease classification with symptom visualization [7], and smartphone-oriented transformer systems [8]. A recent review summarizes the shift toward deeper and transformer-based tomato disease models [9]. At the same time, cassava, wheat, and PlantDoc studies show that performance can change substantially under different backgrounds, devices, viewpoints, and collection protocols [10–12]; Barbedo similarly identified acquisition and dataset factors as important practical limitations [13].

The underlying vision architectures have progressed from ResNet [14] and EfficientNet [15] to Vision Transformers [16]. Vision-language pretraining further extends fixed-set classification by connecting image representations with natural-language supervision, as demonstrated by CLIP [17] and BLIP-2 [18]. Qwen2.5-VL provides a general-purpose multimodal architecture capable of instruction-following visual recognition [19].

Full-parameter adaptation of such models is expensive. LoRA learns low-rank updates while freezing pretrained weights [20], and QLoRA combines LoRA with 4-bit quantization to reduce memory requirements [21]. Kaur et al. recently combined BLIP-2 and LoRA with detection and segmentation for tomato leaf disease recognition [22]. In parallel, distribution-shift studies such as ImageNet re-evaluation [23] and WILDS [24] show that strong in-distribution performance does not guarantee robustness to independently collected data.

This study therefore asks how effectively Qwen2.5-VL-7B-Instruct can be adapted to an 11-class TIH tomato image dataset using QLoRA on a single 24-GB GPU, and how well the selected adapter transfers to an independent labelled benchmark. No new VLM architecture is proposed; the contribution is a reproducible training and evaluation protocol combining data-integrity controls, parameter-efficient adaptation, imbalance-aware sampling, answer-only supervision, validation-based checkpoint selection, a frozen internal test, and post-training external evaluation.

The main contributions are as follows:

- A reproducible QLoRA adaptation pipeline for Qwen2.5-VL-7B-Instruct across 11 tomato condition classes using a single NVIDIA RTX 4500 Ada 24-GB GPU.
- A data-integrity protocol based on image verification, SHA-256 exact deduplication, rejection of contradictory cross-class duplicates, deterministic stratified splitting, and frozen split identities.
- An imbalance-aware training procedure using inverse-square-root weighted sampling and answer-only supervision, while keeping the vision tower frozen and adapting only language-model attention projections.

– A two-stage evaluation protocol consisting of a 649-image internal held-out test and a separate 176-image external benchmark, quantifying both in-domain performance and the remaining cross-domain robustness gap.

2 Materials and Methods

2.1 Experimental Overview

Figure 1 summarizes the complete pipeline. The raw TIH dataset is first audited and deduplicated. The retained images are deterministically divided into frozen training, validation, and test splits. Each image is paired with a fixed class-constrained instruction and its class label. Qwen2.5-VL-7B-Instruct is loaded with 4-bit quantization; the vision tower remains frozen and LoRA adapters are attached only to selected language-model projections. Training uses class-balanced sampling, answer-only token loss, epoch validation, and checkpoint selection by validation macro F1. After the final adapter is selected, it is evaluated once on the frozen internal test split and subsequently on a separate external benchmark.

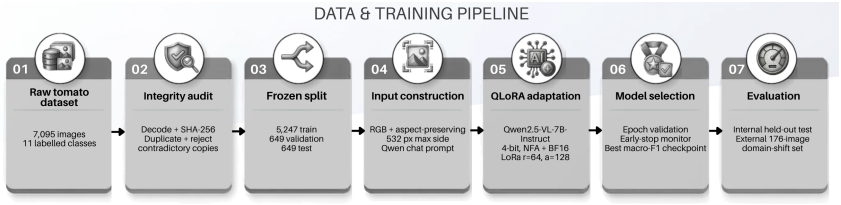


Fig. 1. End-to-end tomato image fine-tuning and evaluation pipeline.

2.2 Private Tomato Dataset and Class Taxonomy

The private tomato dataset initially contained 7,095 labelled images arranged in 11 class folders. The class set used throughout training and internal evaluation was Aphid, Early Blight Fruit, Fruit Borer, Healthy Fruit, Healthy Leaf, Late Blight, Leaf Curl, Leaf Miner, Mealy Bugs, Target Leaf Spot, and White Fly.

The dataset is an image-classification collection spanning healthy tomato material, disease categories, and pest-related conditions. It includes both leaf- and fruit-oriented classes together with healthy reference classes, making the task a fine-grained multi-class recognition problem. After exact duplicate removal and exclusion of contradictory cross-class duplicates, 6,545 usable images were retained. Late Blight was the largest class and Healthy Fruit the smallest, as summarized in Table 1.

Each file was checked for a supported image extension and verified as a decodable image. The raw bytes of every valid image were hashed with SHA-256. Byte-identical images carrying the same class were collapsed to a single representative. If an identical hash appeared under more than one class label, the entire hash group was treated as contradictory and removed rather than forcing a label. In total, 6,545 unique, non-ambiguous images remained, corresponding to 92.25% of the raw collection.

The retained dataset was split using an 80/10/10 target ratio with split seed 20260911. Splitting was class-stratified and deterministic: within each class, ordering was derived from the seed and image hash rather than filesystem order. The resulting frozen split contained 5,247 training, 649 validation, and 649 test images. Table 1 gives the per-class distribution. Within the training split, the largest class was approximately 20.0 times the smallest, motivating imbalance-aware training.

Table 1. Class distribution after data integrity filtering and deterministic splitting.

Class	Train	Validation	Test	Total
Aphid	551	68	68	687
Early Blight Fruit	224	27	27	278
Fruit Borer	346	43	43	432
Healthy Fruit	78	9	9	96
Healthy Leaf	319	39	39	397
Late Blight	1561	195	195	1951
Leaf Curl	424	52	52	528
Leaf Miner	767	95	95	957
Mealy Bugs	320	40	40	400
Target Leaf Spot	264	33	33	330
White Fly	393	48	48	489
Total	5247	649	649	6545

2.3 Image Preprocessing and Prompt Construction

Every image was opened as RGB and resized while preserving its aspect ratio. The longer side was bounded by 532 pixels, and each resized dimension was rounded to a multiple of 28 pixels. Bicubic interpolation was used. No synthetic image generation or external training images were introduced.

Training was formulated as constrained image-to-label generation rather than as a newly added classification head. The fixed user instruction was:

> Classify this tomato plant image into exactly one of these categories: Aphid, Early Blight Fruit, Fruit Borer, Healthy Fruit, Healthy Leaf, Late Blight, Leaf Curl, Leaf Miner, Mealy Bugs, Target Leaf Spot, White Fly. Return only the category name.

The prompt consisted of one image input and this instruction, formatted with the Qwen chat template. The assistant target contained only the exact class label followed by the model's end-of-message token. To prevent the model from being optimized to reproduce the prompt, the collator tokenized the prompt alone and the prompt-plus-answer sequence, measured their common token prefix, and assigned the ignore index -100 to all prompt positions. Consequently, the cross-entropy loss was computed only over target answer tokens. The maximum sequence length was 1,024 tokens.

2.4 Qwen2.5-VL QLoRA Configuration

The base model was Qwen2.5-VL-7B-Instruct [19]. It was loaded with QLoRA-style 4-bit NF4 quantization, double quantization enabled, and BF16 computation. The base weights remained frozen. The visual tower was explicitly excluded from LoRA targeting.

Adapters were attached only to language-model linear projections with suffixes `q_proj`, `k_proj`, `v_proj`, and `o_proj`. LoRA rank was 64, scaling alpha was 128, dropout was 0.05, and no LoRA bias terms were trained.

This configuration introduced 40,370,176 trainable LoRA parameters while all pre-trained model weights remained frozen, corresponding to approximately 0.49% relative to the 8.29-billion-parameter base vision-language model.

Table 2. Final QLoRA training configuration.

Component	Setting	Component	Setting
Base model	Qwen2.5-VL-7B-Instruct	Optimizer	Paged AdamW 8-bit
Quantization	4-bit NF4, double quant.	Learning rate	4×10^{-5}
Compute dtype	BF16	Weight decay	0
Maximum image side	532 px, patch-aligned	LR schedule	Cosine, 3% warmup
Maximum sequence length	1024 tokens	Micro-batch / accum.	1 / 8
LoRA targets	q/k/v/o projections	Effective batch size	8
LoRA r / alpha / dropout	64 / 128 / 0.05	Gradient clipping	1.0
Vision tower	Frozen	Grad. checkpointing	Enabled
		Planned epochs / seed	3 / 20260911

2.5 Imbalance-Aware Sampling

Because the class distribution was highly imbalanced, training did not sample rows uniformly. If class c contained n_c training examples, each image in that class received sampling weight

$$w_c = \frac{1}{\sqrt{n_c}}. \tag{1}$$

A weighted sampler drew a training set of the original length with replacement. The square-root correction was selected to reduce majority-class dominance without forcing full inverse-frequency balancing, which would repeatedly expose very small classes such as Healthy Fruit. The sampler was deterministic under the experiment seed and epoch number. Validation and test distributions were not reweighted.

2.6 Optimization, Validation, and Checkpoint Selection

Training used the Hugging Face Trainer with per-device batch size 1 and eight-step gradient accumulation, giving an effective batch size of 8. Optimization used paged AdamW 8-bit with a learning rate of 4×10^{-5}, 3% warmup, cosine decay, no weight decay, and gradient norm clipping at 1.0. Grad. checkpointing was enabled with non-reentrant checkpointing. BF16 capability was required by preflight checks.

Validation was performed at the end of each epoch. In addition to token-level validation loss, the model generated class labels for the full validation split, and accuracy and macro F1 were computed from parsed predictions. An early-stopping callback with patience 1 monitored validation loss. After training, all validated checkpoints were ranked

primarily by validation macro F1, then validation accuracy, and finally lower validation loss; the highest-ranked checkpoint was reloaded and saved as the final adapter. This separates the early-stopping signal from the final class-balanced model-selection criterion.

Independently of early stopping, the training callback contained a three-hour wall-clock safeguard that requested a final evaluation and checkpoint save once the limit was reached. The observed run ended under this safeguard at 1,908 optimizer steps, corresponding to 2.909 of the planned 3 epochs; the termination is therefore not attributed to the early-stopping callback. Checkpoint metadata included optimizer, scheduler, random-number-generator state, exact split hashes, configuration, and implementation fingerprint to support compatible resumption and prevent accidental mixing of run states.

2.7 Evaluation Protocol

The validation split was used only for training-time monitoring and checkpoint selection. The 649-image internal test split remained frozen and was evaluated after the final adapter had been selected. The test evaluator generated one label per image, mapped the generation back to the 11 canonical class identifiers, and computed accuracy, macro precision, macro recall, macro F1, weighted F1, class-wise precision/recall/F1, confusion matrices, unparseable outputs, and latency. A test identity tied the adapter hash to the frozen test split to discourage repeated test-set tuning.

A second evaluation was performed on an independently stored labelled set of 176 images. Ten of the eleven training classes were represented; Healthy Fruit was absent. The external folder named `Healthy` was mapped to Healthy Leaf. Non-class result folders named `correct` and `wrong` were ignored. This benchmark was inference-only: model gradients were disabled, the model was placed in evaluation mode, and inference ran under `torch.inference_mode()`. No external image was used for training or checkpoint selection.

Because the dataset is imbalanced, macro F1 is treated as the primary aggregate metric. Accuracy and weighted F1 are also reported. Only metrics recorded by the frozen evaluation pipeline are reported.

3 Experimental Setup

3.1 Compute Environment

The final run used one NVIDIA RTX 4500 Ada Generation GPU. Preflight recorded 23.99 GB GPU memory, CUDA 12.4, BF16 support, Python 3.10, PyTorch 2.6.0+cu124, Transformers 5.16.1, PEFT 0.20.0, Accelerate 1.15.0, and bitsandbytes 0.50.2. No distributed-training backend was used.

3.2 Reproducibility and Run Controls

A QLoRA smoke test first exercised the training path and verified adapter writing and reloading. Deep preflight then checked the software environment, CUDA/BF16

capability, class mapping, image inventory and formats, dataset isolation, split counts and hashes, cached base model, processor construction, and evaluation path. Training started only after these checks passed.

Run and split seeds were both 20260911. Training used batch size 1 with eight gradient-accumulation steps; validation used batch size 1. Validation and checkpoint saving occurred at each epoch, with at most three checkpoints retained. Generation was limited to 24 new tokens. Validation loss drove the early-stopping callback, while the final adapter was selected from validated checkpoints by macro F1, then accuracy, then lower validation loss. The independent three-hour wall-clock safeguard ended optimization at 1,908 steps. Internal testing occurred only after final-adapter selection, and external evaluation followed in inference mode without parameter updates.

4 Results

4.1 Training Efficiency

The final run completed 1,908 optimizer updates in 11,562.6 s (192.7 min). Peak reserved GPU memory was 10.57 GB. The final logged training loss was 8.12×10^{-5}. Adapter saving and reloading were verified before evaluation.

The very low final training loss alone would not be evidence of generalization. Model quality is therefore assessed using the held-out internal test and the independent external benchmark, the latter showing a substantial distribution-dependent performance drop.

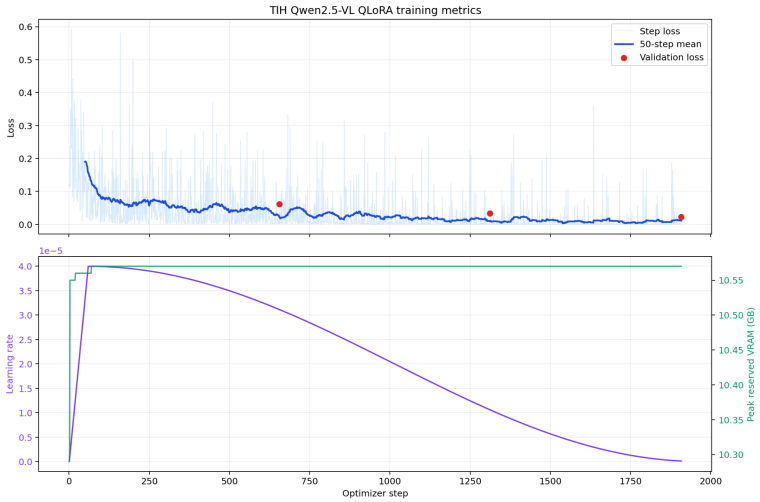


Fig. 2. Training dynamics of the final TIH Qwen2.5-VL QLoRA run, showing recorded step loss, the 50-step moving mean, validation loss, the learning-rate schedule, and peak reserved GPU memory.

4.2 Internal Held-Out Test Performance

On the frozen 649-image internal test set, the final adapter correctly classified 602 images, corresponding to 92.76% accuracy. Macro precision was 92.71%, macro recall

95.57%, macro F1 93.78%, and weighted F1 92.86%. No prediction was unparseable, and mean inference latency was 0.327 s/image.

Class-wise F1 ranged from 81.48% for Target Leaf Spot to 100% for Fruit Borer and Healthy Fruit. Late Blight, the largest class, reached 90.91% F1 with 84.62% recall. Its dominant errors were predictions as Target Leaf Spot (13 images), Leaf Curl (8), and Leaf Miner (7). Figure 3 shows the complete internal confusion matrix.

4.3 External Domain-Shift Benchmark

The independent external benchmark contained 176 labelled images spanning 10 mapped classes. The adapter correctly classified 122 images, producing 69.32% accuracy, 76.08% macro precision, 71.84% macro recall, 71.95% macro F1, and 69.38% weighted F1. Mean latency was 0.373 s/image, and again no generated prediction was unparseable.

Table 3. Aggregate internal and external evaluation results.

Metric	Internal test	External benchmark
Images / classes	649 / 11	176 / 10
Accuracy	92.76%	69.32%
Macro precision	92.71%	76.08%
Macro recall	95.57%	71.84%
Macro F1	93.78%	71.95%
Weighted F1	92.86%	69.38%
Unparseable predictions	0	0
Mean latency	0.327 s/image	0.373 s/image

The external result is 23.44 percentage points lower in accuracy and 21.83 points lower in macro F1 than the internal test. This gap indicates substantial sensitivity to the external acquisition distribution. Late Blight was the clearest failure mode: recall fell to 10% and F1 to 14.29%, with 11 of 20 Late Blight images predicted as Target Leaf Spot. By contrast, Mealy Bugs and White Fly retained F1 values of 93.33% and 91.67%, respectively. Figure 4 visualizes the external errors, while Fig. 5 compares class-wise F1 across the two benchmarks.

4.4 Class-Wise Performance

Table 4 summarizes class-level precision, recall, and F1 for the internal test together with external F1 where available. The external benchmark confirms that aggregate performance degradation is not uniform across classes. Classes such as Mealy Bugs and White Fly transfer relatively well, whereas Late Blight exhibits a pronounced shift. This pattern is more informative than the overall accuracy alone because it identifies which visual distinctions are sensitive to changes in image source or presentation.

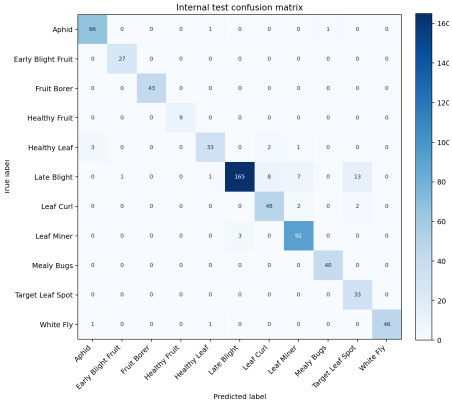


Fig. 3. Confusion matrix on the frozen internal TIH test split.

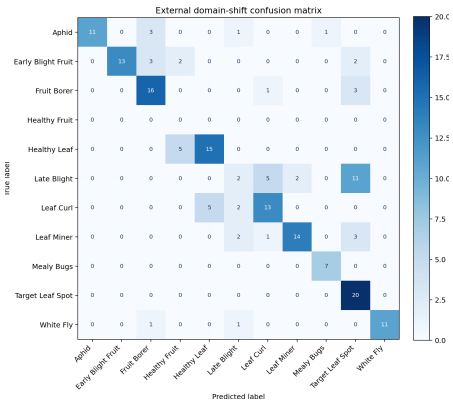


Fig. 4. Confusion matrix on the external domain-shift benchmark. Healthy Fruit is absent from this benchmark.

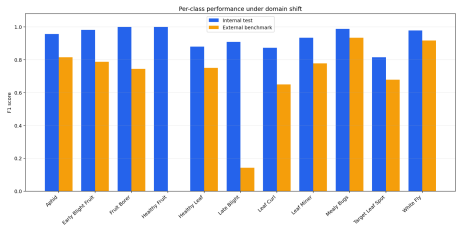


Fig. 5. Per-class F1 comparison between the internal test and external benchmark.

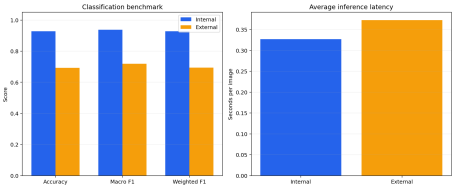


Fig. 6. Aggregate internal and external benchmark performance and average inference latency.

Table 4. Per-class internal metrics and external F1. A dash indicates that the class was absent from the external benchmark.

Class	Support	Prec.	Recall	Int. F1	Ext. F1
Aphid	68	94.29	97.06	95.65	81.48
Early Blight Fruit	27	96.43	100.00	98.18	78.79
Fruit Borer	43	100.00	100.00	100.00	74.42
Healthy Fruit	9	100.00	100.00	100.00	–
Healthy Leaf	39	91.67	84.62	88.00	75.00
Late Blight	195	98.21	84.62	90.91	14.29
Leaf Curl	52	82.76	92.31	87.27	65.00
Leaf Miner	95	90.20	96.84	93.40	77.78
Mealy Bugs	40	97.56	100.00	98.77	93.33
Target Leaf Spot	33	68.75	100.00	81.48	67.80
White Fly	48	100.00	95.83	97.87	91.67

5 Discussion

5.1 Accuracy and Parameter Efficiency

The internal results show that a general-purpose 7B VLM can be adapted to an 11-class tomato task without full-parameter fine-tuning. Only approximately 0.49% of the base model parameters were represented by trainable LoRA parameters, yet the model achieved 93.78% macro F1 on the held-out internal test. The complete run used 10.57 GB peak reserved GPU memory and approximately 3.21 h on a single 24-GB workstation GPU, consistent with the efficiency motivation of LoRA and QLoRA [20, 21]. Classification was expressed directly through the VLM generative interface rather than a new softmax head; constrained prompting and answer-only loss produced zero unparseable generations in both evaluations.

5.2 Effect of Class Imbalance

The training set was highly imbalanced, with Late Blight the largest class and Healthy Fruit the smallest. Inverse-square-root sampling applied a mild correction rather than full inverse-frequency balancing. Minority-class internal performance remained strong, although small supports—especially nine Healthy Fruit test images—mean these estimates should not be interpreted as perfect real-world recognition. Target Leaf Spot showed a different limitation: 100% internal recall but 68.75% precision, with several Late Blight images assigned to Target Leaf Spot, indicating that balancing alone does not resolve fine-grained visual overlap.

5.3 Domain Shift as the Primary Limitation

The central finding is the internal-to-external generalization gap. Accuracy decreased from 92.76% to 69.32% and macro F1 from 93.78% to 71.95%. Late Blight was most sensitive: internal F1 was 90.91%, but external F1 was 14.29%, with 11 of 20 external Late Blight images predicted as Target Leaf Spot. Mealy Bugs and White Fly transferred substantially better, showing that the shift is class dependent rather than uniform. This pattern is consistent with prior evidence that strong controlled-dataset accuracy does not guarantee robustness under changes in background, illumination, capture device, symptom severity, or field conditions [1, 10–13, 23, 24, 9]. Future work should prioritize independently acquired training data and explicit domain-robustness evaluation while keeping the external benchmark untouched.

5.4 Threats to Validity and Limitations

The TIH dataset is private and its acquisition protocol is not fully characterized, limiting direct comparison with public benchmarks. Exact-byte deduplication removes identical files but not necessarily near-duplicate photographs. The external benchmark contains only 176 images across 10 classes; Healthy Fruit is absent and the generic `Healthy` folder is mapped to Healthy Leaf. Hyperparameters were selected empirically on the available hardware rather than through a large factorial search, and broader claims

would benefit from repeated seeds and additional independent datasets. Finally, the study addresses image-level classification only; it does not localize lesions, estimate severity, or recommend treatment.

6 Conclusion

This study presented a complete, resource-efficient pipeline for fine-tuning Qwen2.5-VL-7B-Instruct on the 11-class TIH tomato image dataset using QLoRA. The pipeline combined verified image loading, SHA-256 deduplication, rejection of contradictory duplicate labels, deterministic stratified splitting, patch-aligned image preprocessing, constrained image-to-label prompting, answer-only token supervision, mild class-balanced sampling, validation-based checkpoint selection, and a frozen internal test protocol. The adaptation introduced 40.37 million trainable LoRA parameters, approximately 0.49% relative to the 8.29-billion-parameter base model, while the pretrained weights remained frozen.

The final adapter achieved 92.76% internal accuracy and 93.78% macro F1 with zero unparseable predictions. However, the independent 176-image benchmark produced 69.32% accuracy and 71.95% macro F1, revealing substantial cross-domain sensitivity, especially for Late Blight. The results therefore support two conclusions: parameter-efficient VLM adaptation can deliver strong in-domain tomato disease classification on workstation-class hardware, but robust deployment requires evaluation and training data that cover acquisition conditions beyond the original dataset distribution.

Data and Code Availability

The TIH image dataset used for training is private and is not redistributed with this manuscript. Availability is subject to the data provider's permissions. The reproducible training and evaluation implementation, aggregate metrics, and publication-safe figures are maintained in the associated experiment repository; the trained checkpoint and per-image private predictions are excluded from public distribution.

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